

Page 1 - AWS Cloud Computing Architecture

AWS is a cloud computing platform that provides services such as EC2, S3, RDS and VPC. These services allow organizations to build and deploy applications without managing physical infrastructure.

Overview of Core AWS Services:

- **Cloud computing**: The on-demand delivery of IT resources over the internet with pay-as-you-go pricing.
- **EC2** (Elastic Compute Cloud): Provides secure, resizable compute capacity in the cloud for running application servers.
- **S3** (Simple Storage Service): An object storage service offering industry-leading scalability, data availability, and security.
- **RDS** (Relational Database Service): Makes it easy to set up, operate, and scale relational databases such as PostgreSQL, MySQL, and Oracle.
- **VPC** (Virtual Private Cloud): Enables you to launch AWS resources into a virtual network that you've defined with isolated subnets.

Page 2 - Security, IAM, and Scalability in AWS

AWS Identity and Access Management (IAM) enables administrators to securely manage access to AWS services and resources. Through IAM policies, roles, and multi-factor authentication, organizations maintain least-privilege security controls across cloud architectures.

Auto Scaling and Elastic Load Balancing automatically distribute incoming application traffic across multiple Amazon EC2 instances, ensuring fault tolerance and high availability under variable workloads.